

Có nhiều App

InternalExternalStorage

</> buttonround.xml

```
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <corners android:radius="70dp"/>
    <gradient android:startColor="#E91E63"
        android:endColor="@color/purple_200"
        android:angle="200"
    />
</shape>
```

write internal storage

```
android:background="@drawable/buttonround"
```

```
public class MainActivity extends AppCompatActivity implements View.OnClickListener
```

```
    findViewById(R.id.btnwriteIS).setOnClickListener(this);
    findViewById(R.id.btnreadIS).setOnClickListener(this);
    findViewById(R.id.btnwriteFC).setOnClickListener(this);
    findViewById(R.id.btnreadFC).setOnClickListener(this);
    findViewById(R.id.btnwriteES).setOnClickListener(this);
    findViewById(R.id.btnreadES).setOnClickListener(this);
```

@Override

```
public void onClick(View view) {
```

```
    switch (view.getId()) {
```

```
        case R.id.btnreadIS:
```

```
            String readDataIS = readIS(filename: "myfile.txt");
```

```
            Toast.makeText(context: this, text: "Nội dung file:"
                + readDataIS, Toast.LENGTH_SHORT).show();
```

```
            break;
```

```
        case R.id.btnwriteIS:
```

```
            writeIS(filename: "myfile.txt", data: "Xin chào các bạn!");
```

```
            break;
```

@Override

```
public void onClick(View view) {  
    switch (view.getId()) {  
        case R.id.btnreadIS:  
            String readDataIS = readIS( filename: "myfile.txt");  
            Toast.makeText( context: this, text: "Nội dung file:"  
                + readDataIS, Toast.LENGTH_SHORT).show();  
            break;  
        case R.id.btnwriteIS:  
            writeIS( filename: "myfile.txt", data: "Xin chào các bạn!");  
            break;  
        case R.id.btnreadFC:  
            String readDataFC = readFC( filename: "mycached.cache");  
            Toast.makeText( context: this, text: "Data cache: " +  
                readDataFC, Toast.LENGTH_SHORT).show();  
            break;  
    }  
}
```

```
private void writeIS(String filename, String data) { 1 usage  
  
    try {  
        FileOutputStream fout = openFileOutput(filename, MODE_PRIVATE);  
        fout.write(data.getBytes(StandardCharsets.UTF_8));  
        fout.close();  
        Toast.makeText( context: this, text: "Dữ liệu đã được ghi", Toast.LENGTH_SHORT).show();  
    } catch ( Exception ex )  
    {  
        Toast.makeText( context: this, text: "Error:" + ex.getMessage(), Toast.LENGTH_SHORT).show();  
        ex.printStackTrace();  
    }  
}
```

```
private void writeFC(String filename, String data) { 1 usage  
    File cacheDir = getCacheDir();  
    try (FileOutputStream fout = new FileOutputStream(new File(cacheDir, filename))) {  
        fout.write(data.getBytes(StandardCharsets.UTF_8));  
        Toast.makeText( context: this, text: "Dữ liệu lưu vào cache thành công!",  
            Toast.LENGTH_SHORT).show();  
    } catch ( Exception ex )  
    {  
        ex.printStackTrace();  
        Toast.makeText( context: this, text: "Error:" + ex.getMessage(), Toast.LENGTH_SHORT).show();  
    }  
}
```

```
case R.id.btnwriteFC:  
    writeFC( filename: "mycached.cache", data: "Dữ liệu lưu vào cahce");  
    break;  
case R.id.btnwriteES:  
    writeES( filename: "mysdcard.txt", data: "Dữ liệu lưu vào thẻ nhớ ngoài");  
    break;  
case R.id.btnreadES:  
    String readDataEX = readES( filename: "mysdcard.txt");  
    Toast.makeText( context: this, text: "Data sdcard: " +  
        readDataEX, Toast.LENGTH_SHORT).show();  
    break;  
}
```

```
private String readIS(String filename) { 1 usage  
    try (FileInputStream fin = openFileInput(filename))  
    {  
        byte[] buffer = new byte[1024];  
        int length = 0;  
        length = fin.read(buffer);  
        return new String(buffer, offset: 0, length);  
    } catch ( Exception ex ) {  
        ex.printStackTrace();  
    }  
    return null;  
}
```

```
private String readFC(String filename) { 1 usage  
    File cacheDir = getCacheDir();  
    byte[] buffer = new byte[1024];  
    int length = 0;  
    try (FileInputStream fin= new FileInputStream(new File(cacheDir, filename))) {  
        length = fin.read(buffer);  
        return new String(buffer, offset: 0, length);  
    } catch ( Exception ex ) {  
        ex.printStackTrace();  
        Toast.makeText( context: this, text: "Error: " +  
            ex.getMessage(), Toast.LENGTH_SHORT).show();  
    }  
    return null;  
}
```

```
private void writeES(String filename, String data) { 1 usage
```

```
String sdcard = getExternalFilesDir( type: "" ) + "/" + filename;
```

```
try {
```

```
    OutputStreamWriter writer = new OutputStreamWriter(new FileOutputStream(sdcard));
```

```
    writer.write(data);
```

```
    writer.close();
```

```
    Toast.makeText( context: this, text: "Lưu thành công", Toast.LENGTH_SHORT ).show();
```

```
} catch ( Exception ex )
```

```
{
```

```
    ex.printStackTrace();
```

```
}
```

```
private String readES(String filename) { 1 usage
```

```
String sdcard = getExternalFilesDir( type: "" ) + "/" + filename;
```

```
try {
```

```
    Scanner scan = new Scanner(new File(sdcard));
```

```
    String data="";
```

```
    while (scan.hasNext()) {
```

```
        data = data + scan.nextLine() + "\n";
```

```
    }
```

```
    scan.close();
```

```
    return data;
```

```
} catch ( Exception ex )
```

```
{
```

```
    ex.printStackTrace();
```

```
}
```

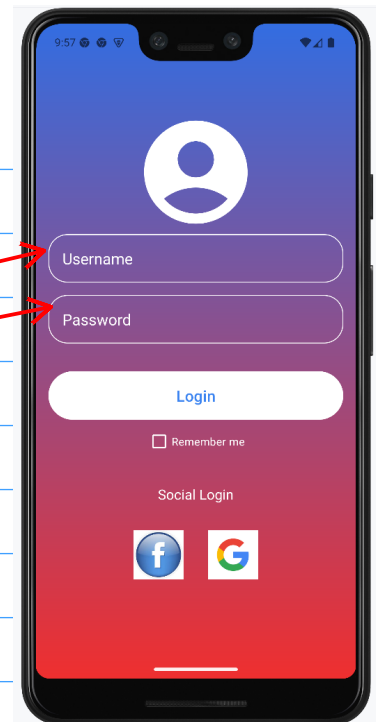
```
app:backgroundTint="@null"
```

</> round_conner.xml

```
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <corners android:radius="24dp" />
    <stroke android:width="1dp"
        android:color="@color/white" />
</shape>
```

</> login_background.xml

```
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <gradient android:type="linear" android:angle="100"
        android:startColor="#336De1"
        android:endColor="#EF2f2f" />
</shape>
```



```
btnLogin.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        //Toast.makeText(MainActivity.this,username.getText().toString().equals("admin") + "",Toast.LENGTH_SHORT).show();
        if(username.getText().toString().equals("admin") && password.getText().toString().equals("123456"))
        {
            if (rememberMe.isChecked())
            {
                SharedPreferences sharedPreferences = getSharedPreferences("login_check",MODE_PRIVATE);
                SharedPreferences.Editor editor = sharedPreferences.edit();
                editor.putString("login", "true");
                editor.apply();
            }
            Intent intent = new Intent(packageContext: MainActivity.this,MainActivity2.class);
            startActivity(intent);
        }
        else {
            Toast.makeText(context: MainActivity.this, text: "Vui lòng nhập lại mật khẩu",Toast.LENGTH_SHORT).show();
        }
    }
})
```

```
private void checkLogin() 1 usage
{
    SharedPreferences sharedPreferences = getSharedPreferences("login_check",MODE_PRIVATE);
    String login = sharedPreferences.getString("login", "");
    if(login.equals("true"))
    {
        Intent intent = new Intent(packageContext: MainActivity.this,MainActivity2.class);
        startActivity(intent);
    }
}
```

```
logout.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View view) {  
        SharedPreferences sharedPreferences = getSharedPreferences( name: "login_check", MODE_PRIVATE);  
        SharedPreferences.Editor editor = sharedPreferences.edit();  
        //editor.putString("login", "false");  
        editor.clear();  
        editor.commit();  
        finish();  
    }  
});
```

